

The On-the-Run Discount

Gao and Klingler

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The opinions expressed here are mine only and do not necessarily reflect the views of the Bank of Canada.

My take-away.

Which story is the important one?

- 1 52-week T-bills become more expensive as they age, an apparent arbitrage — and the reverse of the usual on-the-run pattern.
- 2 A sharp discontinuity near the 26-week reopening. Somehow:
 - ▶ short-covering demand ahead of reopening makes seasoned bills special in repo and rich in cash
 - ▶ the effect disappears after reopening.

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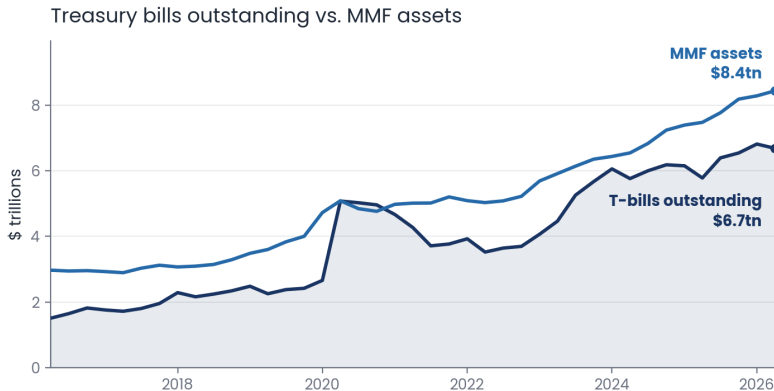
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An emerging topic, and new empirical facts.

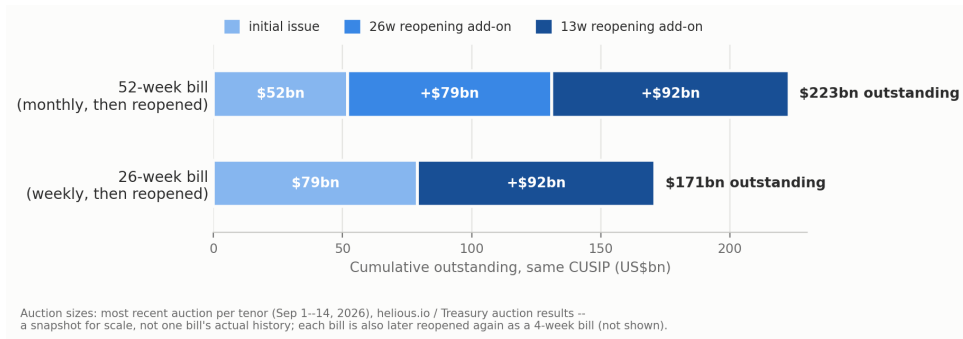
Motivation



Source: Federal Reserve Financial Accounts (Z.1), BOGZ1FL3I3I61110Q and MMMFFAQ027S, quarterly, 2016–2026.

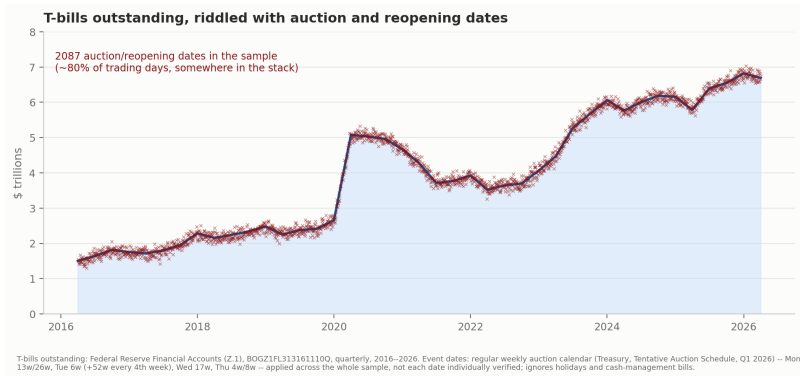
T-bill supply surged since 2015; one part in the broader fiscal story.

Issuances and Reopenings...



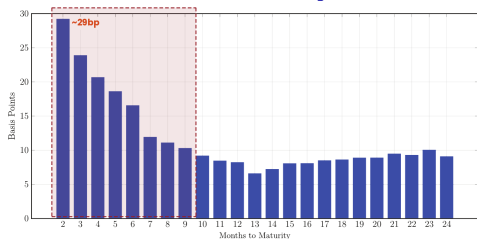
- Bill issuances and reopenings are frequent—contrast with Treasury bond issuances and public equity market issuances.

Identification is not scarce

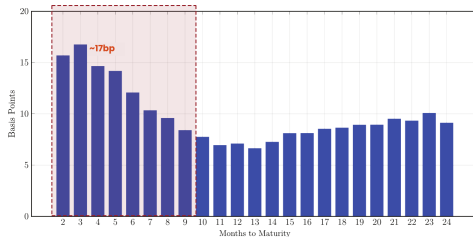


The Treasury bill market is unusual in combining relevance, enormous scale, frequent issuances, and a relatively thin empirical literature.

Treasuries in the money market—Fleckenstein and Longstaff (2024)



Average Richness vs Months to Maturity
All Treasuries



Average Richness vs Months to Maturity
Notes and Bonds only

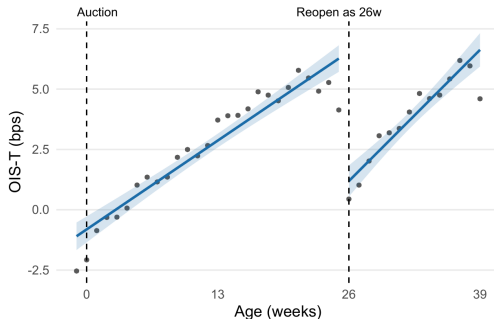
- Richness is risk-free rate minus T-bill yield: $\rho(m) = r(m) - y(m)$, m is maturity.
- MMFs favor shorter maturities (and largely older securities); slope is $\approx 2\text{bp/month}$.
- Mechanism: MMFs target average portfolio maturity below 60 days (Rule 2a-7).
- F&L: target WAM was cut from 90 to 60 days in 2010, which led to a steeper slope afterward.

Zooming on 52-week T-bills—Gao and Klingler (2026)

G&K (Aug. 2026), Figure 3. Track 52-week bills over time.

Dashed lines mark the initial issuance and first reopening.

- Pre-reopening slope $\approx -1\text{bp/month}$ seems in line with F&L evidence.
- But richness drops sharply at the 26-week reopening and resumes rising.
- Was the Fleckenstein and Longstaff (2024) story around MMFs all that there is?
- Or distinct questions: (i) why the arbitrage survives and (ii) why such a large and persistent reopening break (the sawtooth)?



MMF Demand vs Arbitrage: A toy model.

The richness pattern sits between many forces:

1 T-bill supply

- ▶ issue size $Q_s(m)$ and total float $Q = \int_0^{1y} Q_s(m) dm$.

2 MMFs' demand $Q_f(m)$ for money-like assets:

- ▶ Assume flat risk-free term structure $r(m) = r$.
- ▶ $\max_{Q_f(m) \geq 0} \int_0^{1y} [v_0 - \psi m - \rho(m)] Q_f(m) dm \quad \text{s.t.} \quad \int_0^{1y} Q_f(m) dm \leq A$

3 Frictions limit shorts $Q_a(m)$ by arbitrageurs

- ▶ $\max_{Q_a(m)} \rho(m) Q_a(m) - \frac{\gamma}{2} \frac{Q_a(m)^2}{Q_s(m)}$

A toy model: FOCs and equilibrium

MMFs

FOC: $\rho(m) = v_0 - \psi m - \mu$.

Trade off returns and regulatory constraints.

ψ : regulatory costs per unit of maturity;

μ : shadow cost of scarce capacity.

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Trade off profits and short-selling costs.

γ : marginal cost rising with relative position size.

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Market clearing: $\rho(m) = \underbrace{\gamma \left(\frac{A}{Q} - 1 \right)}_{\text{level}} + \underbrace{\psi}_{\text{slope}} (WAM - m)$

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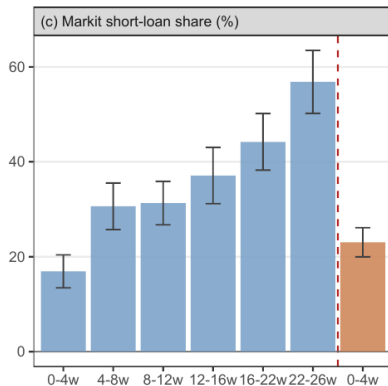
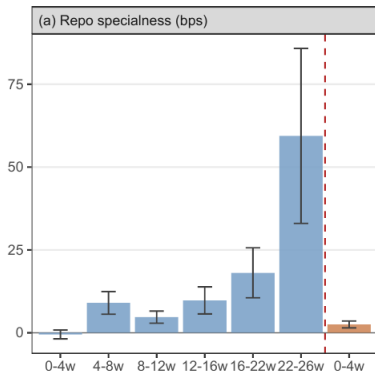
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- Richness is linear in maturity. No sawtooth in an MMF-driven model?
- Maturity composition not a big deal. Adding re-issuances shifts the level—no kinks.

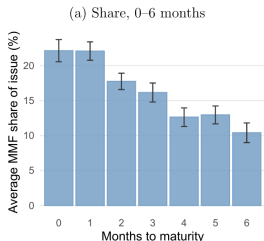
The reopening specialness cycle?



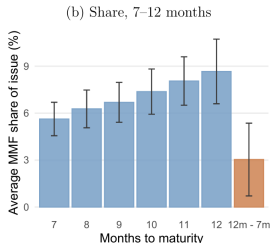
G&K (Sept. 2026), Fig. 2 (a, c): specialness and Markit short-loan share by age; dashed line marks the 26-week reopening.

- Both security lending fees and short positions rise steadily toward the reopening, then fall sharply once it settles.
- Two messages: (i) This explains how MMF-driven richness can survive; (ii) but move the sawtooth puzzle elsewhere?

MMF Holdings



Post-reopening (0–6 months)

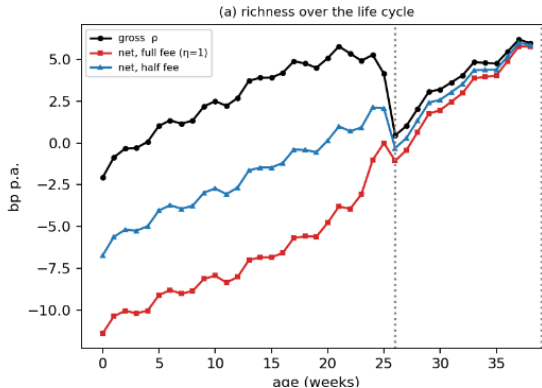


Pre-reopening (7–12 months)

G&K (Sept. 2026), Figure 6 and Figure IA.2 (panel a). The 26-week reopening falls at the boundary between the two panels shown.

- Intuitive: MMFs that do not earn the lending fees should under-weight T-bills nearing the reopening.
- In Vayanos & Weill, the richness is essentially entirely linked to the value of future lending fees that *bond holders* earn.
- How rich are these T-bills *net of the fee revenue*?

Net of the fee — tongue-in-cheek



Speculative, reading results across tables and adding assumptions. See appendix for details. Blame the LLMs if you disagree.

- Richness net of specialness—a purer convenience yield—may look quite different.
- This turns the spotlight on the short-sellers.

One path forward: focus on the MMFs, follow a model.

- Surprise changes to the T-bill issuance schedule should move the level and the slope of convenience yields.
- Surprise shifts into and out of money market funds should move the level of convenience yields.
- Is the Treasury responding to richness, say by varying issuances?

Or turn the emphasis on arbitrageurs, and why they drive an intense level of short-selling activity.

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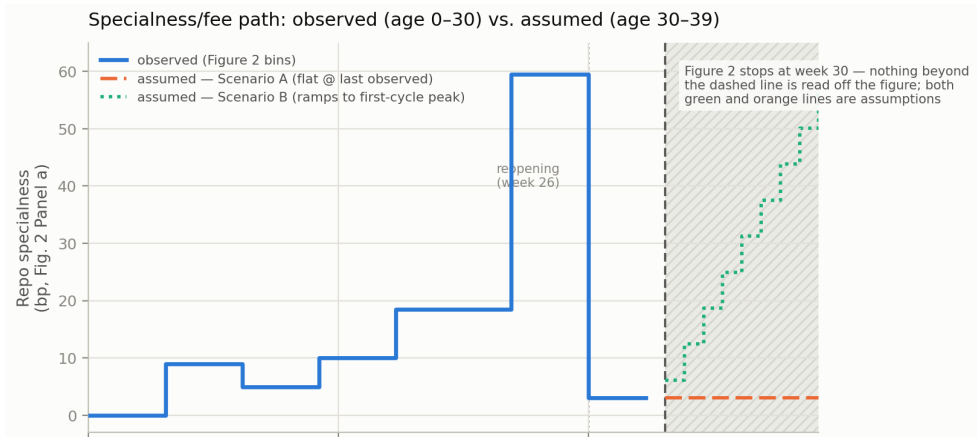
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Conclusion

- An important topic. Understanding the T-bill market is increasingly relevant.
- A literature gap.
- Important results. Which path forward?
 - 1 Build on F&L: the interaction between MMFs' regulatory demand, limits to arbitrage, and variations in Treasury issuances and reopenings.
 - 2 Pivot to the intense short-selling activity, and establish its motives/economic function.

Appendix: the week 30–39 fee assumption



Appendix: gross vs. net richness under both scenarios

